ENVIRONMENTAL PRODUCT DECLARATION

in accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for

FirePro Ei120 Single Glazed Fire Rated Partition System

from

Radii Planet Group Ltd

EPD registration number: EPD-IES-0025725:001
Version date: 2026-03-09
Validity date: 2031-03-08
An EPD may be updated or depublished if conditions change.
To find the latest version of the EPD and to confirm its validity www.environdec.com

Programme: The International EPD® System
Programme operator: EPD International AB
www.environdec.com

EPD type: EPD of multiple products, based on worst-case results, covering:
FirePro Ei120 single glazed mullioned partition system
FirePro Ei120 single glazed butt jointed partition system

Radii Planet Group.

GENERAL INFORMATION

PROGRAMME INFORMATION	
EPD programme	The International EPD® System
EPD programme operator	EPD International AB - Box 21060 - SE-10031 Stockholm - Sweden www.environdec.com - support@environdec.com

PRODUCT CATEGORY RULES (PCR)	
The CEN standard EN 15804:2012 + A2:2019 serves as the core PCR	
Product Category Rules	PCR 2019:14 Construction products (EN 15804+A2) Version 2.0.1 (2025-06-05) - UN CPC4212
PCR review conducted by	The Technical Committee of the International EPD® System See www.environdec.com for a list of members Review chairs: Rob Rouwette (chair), Noa Meron (co-chair) The Review Chairs may be contacted via info@environdec.com

THIRD-PARTY VERIFICATION	
External and independent ("third-party") verification of the declaration and data, according to ISO 14025 via individual EPD verification, without a pre-verified LCA/EPD tool	
Third party individual verifier	Dr Matthew Fishwick, Fishwick Environmental Ltd
Approved by	The International EPD® System
Procedure for data follow-up during EPD validity	involves third party verifier: ☐ yes ■ no

LCA ACCOUNTABILITY
EuGeos Limited – www.eugeos.co.uk - info@eugeos.co.uk

OWNERSHIP AND LIMITATIONS ON USE OF EPD

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall

- *be based on the same PCR (including the same first-digit version number) or be based on fully-aligned PCRs or versions of PCRs;*
- *cover products with identical functions, technical performances and use (e.g. identical declared/functional units);*
- *have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant);*
- *apply identical impact assessment methods (including the same version of characterisation factors);*
- *and be valid at the time of comparison.*

For further information about comparability, see EN 15804 and ISO 14025.

Radii Planet Group.

INFORMATION ABOUT THE EPD OWNER

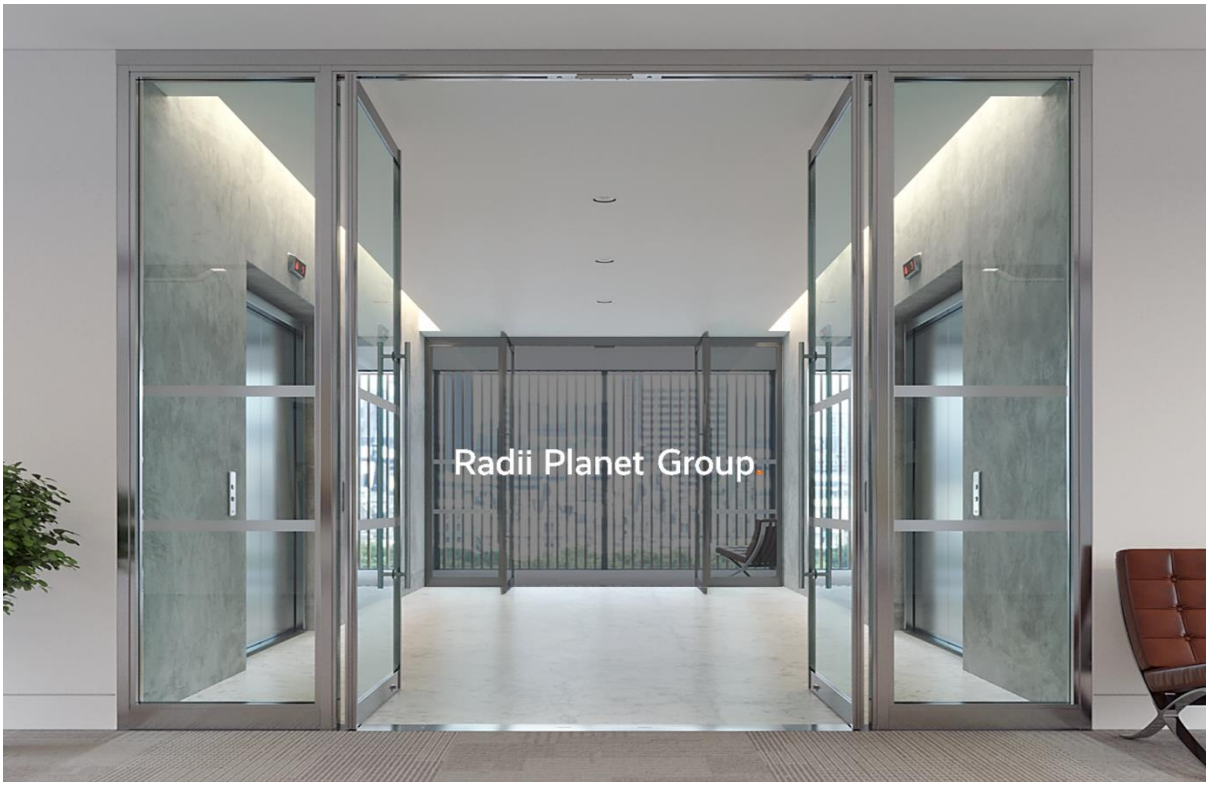
EPD OWNER: Radii Planet Group Ltd

Radii Planet Group Ltd is based throughout the UK with 7 regional contracting offices and manufacturing facilities in Burgess Hill, West Sussex, UK and manufactures a range of partition and door systems.

Utilising a design-led and consultative process, the company specialise in delivering bespoke, adaptable glazing partitioning solutions to produce workspaces that meet all technical and aesthetic requirements.

Radii Planet Group's team of designers and technical specialists work closely with architects and clients to help turn an initial concept into a fully realised design brief, with acoustic performance and aesthetics at the forefront.

Radii Planet Group is certified to ISO 9001, ISO 14001 and ISO 45001.



CONTACT

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PRODUCT INFORMATION

PRODUCT NAME: FirePro Ei120 Single Glazed Fire Rated Partition System

PRODUCT IDENTIFICATION

FirePro single glazed fire rated partition system certified to Ei120, powder coated, mullioned or butt jointed.

PRODUCT DESCRIPTION

The FirePro system is available as a mullioned or butt jointed option and is certified up to Ei120 (integrity and insulation). Suitable for all high-end environments including retail, commercial, public or residential, the system can be finished in a brushed, bright polished or polyester powder coated finish.

Manufactured in the UK using steel and bi-directional fire rated glazing as standard, FirePro ensures safety and compliance without compromising on design. Working closely with our manufacturing partners to ensure minimal waste and machining processes, FirePro offers high-performance fire-rated glazing for any project.

FirePro has been independently tested and achieved Certifire certification. Certifire is an independent third-party passive fire product certification scheme that assures performance, quality, reliability and traceability of products and systems.

This multiple-product EPD is based on a FirePro Ei120 single glazed mullioned partition system with 54mm glass, excluding doors, this configuration having the highest LCA impact results compared to the butt jointed option, also analysed in this EPD. The assumed opening size for the partition is 3x3m.

UN CPC CODE

4212 - Doors, windows and their frames and thresholds for doors, of iron, steel or aluminium under the UN CPC classification system V2.1.

MANUFACTURING

FirePro frames are manufactured to the desired specification at Radii Planet Group's trusted partners in Yorkshire, UK. Intermediate steel products are cut, formed and coated to make the framework, voids being filled with insulation as required to meet the performance criteria. Fire rated glazing is cut to size and delivered to the client site.

MANUFACTURING SITE:

Bradford, Yorkshire, UK

PACKAGING AND TRANSPORTATION

Steel frameworks are delivered on wooden pallets. Glass is delivered to client sites on reusable frames or stillages.

INSTALLATION

Installation is a manual operation only using hand power tools. Framework is attached to walls and ceilings, with glass fitted into place with polymer glazing seals.

PRODUCT USE & MAINTENANCE

The partitions should be cleaned and inspected regularly. Surfaces can be cleaned by wiping with a damp cloth to remove surface dirt. Solvent cleaners must never be used as these will damage the surface finish.

NB: This advice is for information only as the use stage is excluded from the LCA.

END-OF-LIFE

When the partitioning systems are removed without further re-installation intended, the main constituent materials should be separated for recycling.

The following European Waste Catalogue (EWC) codes below apply to the product or its main constituents when removed from the building:

EWC 17 02 02	Glass
EWC 17 04 05	Iron and Steel
EWC 17 09 04	Mixed construction and demolition waste would apply to any material removed from the inside of steel profiles

Disposal of materials and components must be carried out safely with due consideration for the prevailing environmental, health & safety regulations and disposal procedures.

RESIDUAL RISKS & EMERGENCIES

There are no residual risks associated with the normal day to day usage of the partitioning systems, provided they are correctly installed, regularly inspected and maintained.

FURTHER PRODUCT INFORMATION

Detailed product information and datasheets can be found on our website: www.radiiplanetgroup.com or by contacting info@radiiplanetgroup.com

CONTENT DECLARATION

This content declaration is based on the FirePro Ei120 single glazed mullioned partition system with 54mm glass, the worst-case product representing the whole group. It is based on the typical properties and composition of the materials in the product, and shown below per declared unit (DU) of 1m^2; the total DU mass is 120.8kg (including packaging).

Although there is likely to be some post-consumer recycled material present in the steel framework because all steelmaking uses some ferrous scrap as input, in this EPD post-consumer recycled content is reported as zero because the actual content is unknown.

PRODUCT CONTENT				
Material/ substance name	Mass	Post-consumer recycled material	Biogenic material	
	kg	mass – % of product	mass - % of product	kg C/DU *
Glass	104.7	0	0	0
Steel	8.5	0	0	0
Mineral insulation	4.1	0	0	0
Polymers	0.1	0	0	0
Screws/fixings	0.1	0	0	0
Total	**117.5**	**0**	**0**	**0**

PACKAGING MATERIALS			
Material name	Mass kg	Mass % vs product	Biogenic material kg C/DU *
Wood	3.3	3	1.3
Total	**3.3**	**3**	**1.3**

** 1kg biogenic carbon in the product/packaging is equivalent to the uptake of 44/12 kg of CO_2*

The above content declaration is for guidance only and does not constitute a specification.

At the time of data collection, no substance included in the Candidate List of Substances of Very High Concern for authorisation, derived under the REACH Regulations, is present in the product above the threshold for registration with the European Chemicals Agency.

LCA INFORMATION

This section of the EPD records key features of the LCA on which it is based.

LCA SOFTWARE AND DATABASE

The LCA was carried out by EuGeos using openLCA software (V2.5) and production data for calendar year 2023. Background data were taken from the ecoinvent database (v3.10).

DECLARED UNIT (DU)

The declared unit is one square metre of partition system.

AREA DENSITY: 117.5 kg/m^2

REFERENCE SERVICE LIFE

No reference service life is specified in this EPD.

GEOGRAPHICAL SCOPE

Modules A1 - A3 represent production in the UK with basic raw material extraction from unknown sources and therefore assumed global.

Module A4, A5 and C1-C4 represent delivery, installation and end-of-life management in the UK.

LCA SCOPE AND SYSTEM BOUNDARIES

The system boundary of the EPD is defined using the modular approach set out in EN 15804. This is a "type b" EPD: cradle-to-gate with options, modules C1–C4, module D and with optional modules (A1–A3 + C + D and additional modules A4 & A5 - see below). The use stage (modules B) is omitted because the products are passive in use and no data are available on which to build realistic scenarios relating to possible repair or refurbishment. As permitted by EN 15804 and required by the PCR, modules A1-A3 are declared in aggregated form.

Therefore, as well as the core processes, the system includes:

- production of all raw materials and components from basic resources;
- transport of those materials at all stages up to Radii Planet Group's manufacturing facility and transport to site;
- installation in the building; the production of fuels and energy carriers and their delivery to manufacturing sites;
- the treatment of all wastes.

Benefits and loads beyond the system boundaries (Module D) are also considered.

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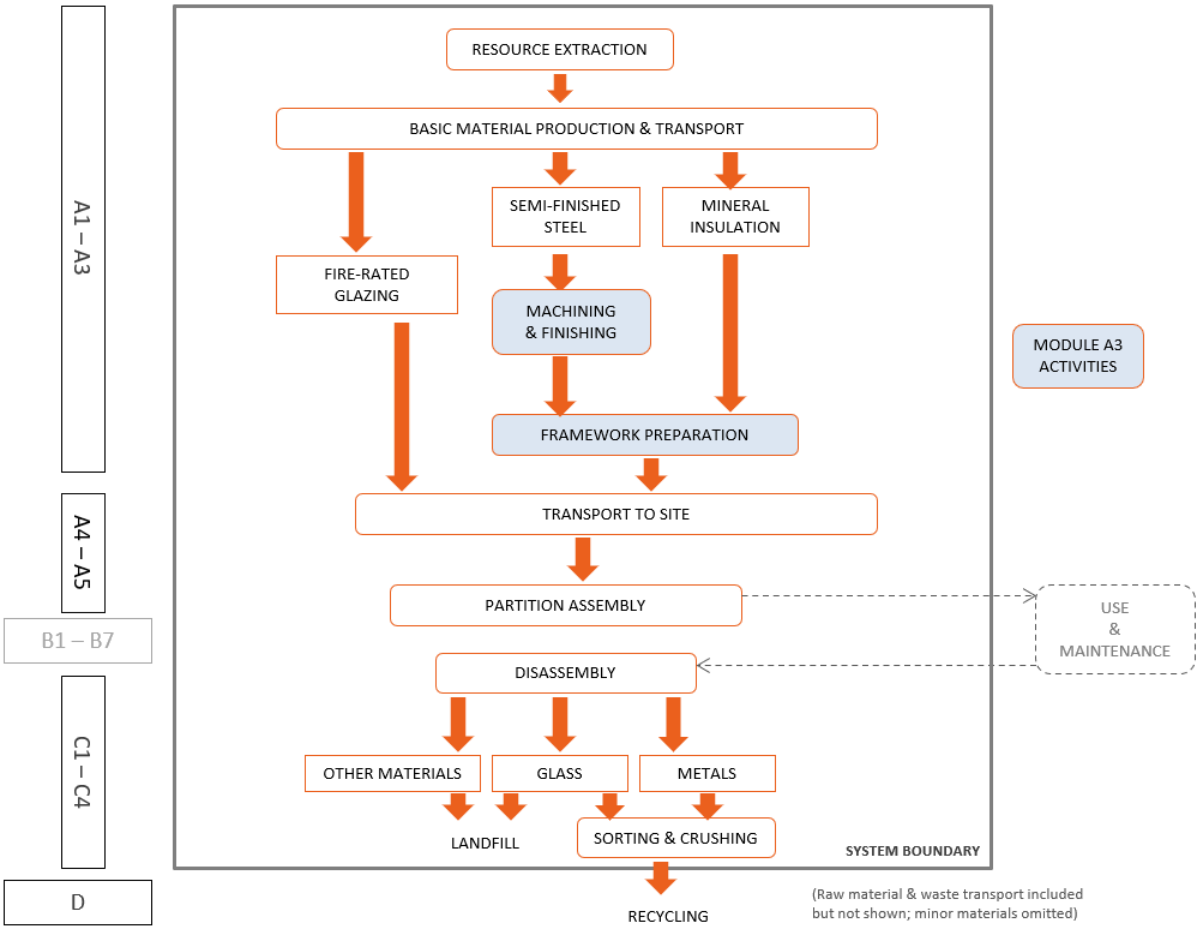
Non-reusable packaging used to deliver products and / or components to the place of installation is included in the LCA. Capital equipment in the foreground system is excluded; it is included in the background data.

PRODUCT STAGE			CONSTRUCTION PROCESS STAGE		USE STAGE							END OF LIFE STAGE				Benefits & loads beyond the system boundaries
Raw material supply	Transport	Manufacturing	Transport to the site	Assembly	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste treatment	Disposal	Reuse- recovery- recycling- potential
A 1	A 2	A 3	A 4	A 5	B1	B2	B3	B4	B5	B 6	B 7	C 1	C 2	C 3	C 4	D
DECLARED MODULES X: included in LCA; ND: module not declared; NR: module not relevant																
X	X	X	X	X	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
GEOGRAPHY *																
GLO		UK	UK	UK	-	-	-	-	-	-	-	UK	UK	UK	UK	UK
SPECIFIC DATA USED *																
4%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
VARIATION – PRODUCTS *																
<10%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
VARIATION – SITES *																
N/A **			-	-	-	-	-	-	-	-	-	-	-	-	-	-

** share of specific data (in GWP-GHG results)*
*** N/A (not applicable): single site*

PROCESS FLOW DIAGRAM

The product life cycle covered by this EPD is illustrated below.



DATA SOURCES & DATA QUALITY

TIME REPRESENTATIVENESS

Data collected for the core processes (production) cover calendar year 2023. The producer-specific data used in LCA calculations are based on 1 year averaged data and have been updated within the last 5 years. These data were checked to ensure that sufficient materials and water are included within the inputs to account for all products, wastes and emissions.

BACKGROUND DATA

Background (generic) data from the ecoinvent database (v3.10) fulfil the EN 15804 requirement that generic data used in the LCA have been updated within the last 10 years. Data quality has been reviewed for processes that contribute significantly to the overall LCA.

Significant datasets are those for steel profile and fire-resistant glass; these are assessed as being very good and good quality respectively.

DATA QUALITY ASSESSMENT

Data quality has been assessed using the data quality level and criteria schemes of the UN Environment Global Guidance on LCA database development (GLAD); the overall data quality ratings for significant datasets (that together account for >80% of the declared environmental impact indicators) are presented with other data quality information in the table below.

The geographical applicability of the different modules is shown in the table in the LCA Scope section of this EPD, and the manufacturing technology indicated under 'Manufacturing' in the Product Information section.

Process	Data source type	Source	Reference year	Data category	Overall data quality	Share of primary data, of GWP-GHG results for A1-A3
Glass manufacture	database	ecoinvent, EuGeos	2023	secondary	good	0%
Steel production	database	ecoinvent	2023	secondary	good	0%
Electricity	database	ecoinvent	2023	primary	very good	<10%
Heat from natural gas	database	ecoinvent	2023	primary	very good	<10%
Other processes	database	ecoinvent	2023	secondary	good	0%
Total share of primary data, of GWP-GHG results for A1-A3						**4%**

The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data to increase the representativeness of and comparability between EPD. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

Data representing seals and liners are assessed as being poor due to limited information about the composition of these items. These constituents account for <1% of any product and thus meet the cut-off criteria – the use of poor or proxy data to represent them is justified on the basis of their low relevance to the overall assessment.

CUT-OFF CRITERIA

The collected data covered all raw materials, consumables and packaging materials; associated transport to manufacturing sites; process energy and water use; direct production wastes; emissions to air and water.

According to EN 15804 and the PCR, flows can be omitted (cut-off) from a core process in the LCA up to a maximum of 1% of the total mass of material inputs or 1% of the total energy content of fuels and energy carriers.

ALLOCATION

In the background data, the ecoinvent default allocation is applied to all processes except those in which secondary materials are used, where the "cut-off" allocation is applied. This ensures that secondary materials are free of upstream burdens that arise prior to their reaching the "end of waste" state, in accordance with Section 6.3.4.2 of EN 15804.

Where it cannot be avoided, allocation in primary data is based on physical relationships. No data were available as a basis for allocating burdens to steel scrap generated in the manufacture of framework; the product therefore carries 100% of the burdens of the manufacturing activity. Wastes sent to recycling from C1 are assumed still to be waste until they are separated and crushed; the ferrous scrap and glass cullet generated are modelled secondary materials that have passed the end-of-waste state.

ASSUMPTIONS & ESTIMATES

Inputs to and outputs from the system are accounted for over a 100-year time period, except for biogenic carbon. Long-term emissions are therefore omitted from the impact assessment part of the LCA, except for biogenic carbon releases from waste disposal to which no time cut-off is applied.

Details of polymers and fixings accounting for <0.5% of the finished product were not available. These were included in a simplified way, as pvc extrusions and machined steel parts respectively.

For the calculation of primary energy as materials (PERM, PENRM), the following net calorific values are applied to the quantities of materials and packaging shown in the product's bill of materials:

- wood – 16 MJ/kg
- polyethylene (PE) – 45 MJ/kg
- polypropylene (PP) – 43MJ/kg
- polyvinylbutyral (pvb); polyethyelene terephtahlate & derivatives (PET, PET-G); polyvinyl chloride (pvc) – 20MJ/kg
- other polymers – 30MJ/kg

When accounting for the primary energy indicators PENRM and PERM across the life cycle, the approach described as Option B in PCR 2019:14 is applied.

ELECTRICITY MODELLING

TYPE OF ELECTRICITY

Electricity supplied to operations providing primary data is modelled as the residual mix for 2023. The dataset represents the following mix of generation types at high-voltage level:

Energy source	Share of the mix (%)
Natural gas	50
Nuclear	19
Wind	11
Coal	16
Oil	2
Biomass	2
Climate impact (GWP-GHG)	**0.47kg CO_2e/kWh**

Electricity used in Module A3 accounts for <5% of the overall GWP-GHG value and is of low significance as a driver of the indicators reported in this EPD.

SCENARIO INFORMATION

MODULE A4

Parameters applied for the transport of materials in Module A4 are shown in the table below.

A4 transport scenario parameters	Quantity and unit (per declared unit)
Vehicle type and load capacity	lorry 16 - 32 metric ton EURO5
Fuel type and consumption of heavy goods vehicle used for road freight	diesel, 0.2l/km
Capacity utilisation (including empty returns)	6t / 33%
Distance	330 km
Bulk density of transported products	2500 – 3500 kg m^{-3}

MODULE A5

Installation in the building, Module A5, comprises attachment of the framework to the ceiling, walls and floors and installation of the glass. The framework and glass are all produced to the sizes required for any particular project, therefore there are no material losses. Electricity use of 0.075kWh per declared unit is assumed for hand power tools such as drills and screwdrivers; this is modelled as the consumption mix of UK low voltage electricity from the grid.

Wooden packaging is assumed to become waste in module A5 and to be incinerated with energy recovery. The default distance of 130km is applied for transport to waste management. Energy exported from incineration is included in the Exported Energy indicator and in the Module D calculation.

Fixings and sealants used in installation are omitted.

MODULES C

Module C1:

It is assumed that – because of their size - the products are disassembled in situ rather than transported prior to disassembly, then collected separately from mixed construction waste. Energy use for disassembly is assumed to be the same as energy use for installation.

Module C2:

The default scenario comprises recycling of metals and disposal to landfill of the remaining materials. The default distance to waste treatment other than incineration (from PCR 2019:14 v2.01) is applied: road transport over 80km in a 16-32t Euro5 truck, with a load factor of 50%.

Module C3:

For consistency with the system boundary applied for input of recycled material, sorting and pressing of scrap metal, and crushing of waste glass are included to represent treatment of materials to be recycled in Module C3. For materials sent to landfill, it is assumed that there is no treatment. In the 100% recycling scenario, it is assumed that the whole product, including constituents that cannot be recycled, are subject to this treatment.

Module C4:

Disposal of all materials that are not recycled. In the default scenario, this means all non-metallic materials. In the 100% recycling scenario, the mineral insulant and polymer content are assumed to be unrecyclable and to be disposed of to landfill with other residues of the treatment process in C3.

MODULE D

In Module D, benefits and loads are calculated for the net quantity of material recycled: the recycling rate minus recycled content. In this case, the recycled content is assumed to be zero for all constituents of the product. Module D calculations include wastes recycled and recovered from Module A5.

Electricity and heat are recovered from the incineration of wooden packaging. Energy exported from the incineration is included in Module D, with conversion to electrical energy assumed to be 20% efficient and conversion to thermal efficiency 52%. The potential benefits derived from these exports are calculated on the basis that heat generated substitutes for gas-fired heat generation, and electricity generated substitutes for average UK grid electricity.

The quantities recovered and/or recycled for each of the end-of-life scenarios are shown in the table below.

Module D - scenario parameters							
Output to recycling / recovery	Assumed recycling / recovery	Displaced input flow	Quality factor	Net output			
				Quantity			
				Default scenario	100% landfill scenario	100% recycling scenario	Unit
Steel	recycling via re-melting in electric arc furnace	converter steel	0.85	9	0	9	kg
Glass	recycled	white packaging glass	0.85	0	0	84	kg
Wood from packaging	energy recovery	heat from natural gas	-	28	28	28	MJ
		electricity, grid mix, UK	-	11	11	11	MJ

ENVIRONMENTAL INDICATORS AND CHARACTERISATION METHODS

This EPD contains environmental information in the form of quantitative indicator values for a number of parameters, which encompass calculated environmental impact potentials, resource and energy use, and waste generation. These parameters are listed below along with the abbreviations used for them in the tables of indicator values that follow.

The EF3.1 characterisation factors have been applied when calculating the potential environmental impacts.

Potential Environmental Impacts (mandatory indicators as per EN 15804:2012+A2:2019)	Abbreviation	Unit	Source
Global warming potential - fossil fuels	GWP-fossil	kg CO_2-eq	GWP100, EN 15804. Version 2: EF 3.1, February 2023
Global warming potential - biogenic	GWP-biogenic	kg CO_2-eq	GWP100, EN 15804. Version 2: EF 3.1, February 2023
Global warming potential - land use and land use change	GWP-luluc	kg CO_2-eq	GWP100, EN 15804. Version 2: EF 3.1, February 2023
Global warming potential - total	GWP-total	kg CO_2-eq	GWP100, EN 15804. Version 2: EF 3.1, February 2023
Depletion potential of the stratospheric ozone layer	ODP	kg CFC11-eq	ODP, EN 15804. Version2, February 2023
Acidification potential of land and water	AP	mol H^+ eq	AP, accumulated exceedance, EN 15804. Version 2, February 2023
Eutrophication potential - freshwater	EP-freshwater	kg P eq	EP, aquatic freshwater, EUTREND model, EN 15804. Version 2, February 2023
Eutrophication potential - marine	EP-marine	kg N eq	EP, aquatic marine, EUTREND model EN 15804. Version 2, February 2023
Eutrophication potential - terrestrial	EP-terrestrial	mol N eq	EP, terrestrial, accumulated exceedance, EN 15804. Version 2, February 2023
Formation potential of tropospheric ozone	POCP	kg NMVOC eq	POCP, LOTOS-EUROS as applied in ReCiPe, EN 15804. Version 2, February 2023
Abiotic depletion potential - minerals & metals [1]	ADPMM	kg Sb-eq	ADP minerals & metals, EN 15804. Version 2, February 2023
Abiotic depletion potential - fossil resources [1]	ADPFF	MJ	ADP fossil resources, EN 15804. Version 2, August 2021
Water (user) deprivation potential, deprivation-weighted water consumption [1]	WDP	m^3	Water deprivation (Available Water Remaining - AWARE - method), EN 15804
Potential Environmental Impacts (additional mandatory indicator)	**Abbreviation**	**Unit**	**Source**
Global warming potential - biogenic excluded [2]	GWP-GHG	kg CO_2-eq	GWP100, EN 15804. Version 2: EF 3.1, February 2023, CF CO_2bio = 0
Resource Use Indicators [6]	**Abbreviation**	**Unit**	**Source**
Renewable primary energy as energy carrier	PERE	MJ, ncv [3]	Local calculation, difference between PERT & PERM
Renewable primary energy resources as material utilisation	PERM	MJ, ncv [3]	Local calculation, material NCVs [6]
Total use of renewable primary energy resources	PERT	MJ, ncv [3]	Spindler & Ciroth 2023
Non-renewable primary energy as energy carrier	PENRE	MJ, ncv [3]	Local calculation, difference between PENRT & PENRM
Non-renewable primary energy as material utilisation	PENRM	MJ, ncv [3]	Local calculation, material NCVs [6]
Total use of non-renewable primary energy resources	PENRT	MJ, ncv [3]	Spindler & Ciroth 2023
Use of secondary material	SM	kg	Spindler & Ciroth 2023
Use of renewable secondary fuels	RSF	MJ, ncv [3]	Spindler & Ciroth 2023
Use of non-renewable secondary fuels	NRSF	MJ, ncv [3]	Spindler & Ciroth 2023
Use of net fresh water	FW	m^3	International EPD System, GPI 5 [4]

Waste Production Indicators	Abbreviation	Unit	Source
Hazardous waste disposed	HWD	kg	International EPD System, GPI 5 [4]
Non-hazardous waste disposed	NHWD	kg	International EPD System, GPI 5 [4]
Radioactive waste disposed	TRWD	kg	International EPD System, GPI 5 [4,5]
Output Flows Indicators [6]	**Abbreviation**	**Unit**	**Source**
Components for re-use	CRU	kg	Spindler & Ciroth 2023
Materials for recycling	MFR	kg	Spindler & Ciroth 2023
Materials for energy recovery	MER	kg	Spindler & Ciroth 2023
Exported energy, electricity	EEE	MJ, ncv [3]	Spindler & Ciroth 2023
Exported energy, thermal	EET	MJ, ncv [3]	Spindler & Ciroth 2023

1 - Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator
2 - GWP-GHG indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero
3 - ncv: net calorific value
4 – As detailed at https://www.environdec.com/pcr/env-perf-indic/gpi5
5 – Long-term storage is considered to constitute "treatment"
6 - Additional flows entering or leaving the specific product system under study are included in the calculation of resource and output flow indicators as appropriate. Such additional flows are recorded in the "Assumptions and Estimates" section of this EPD, along with NCV used in local calculations

ENVIRONMENTAL PERFORMANCE

INDICATOR RESULTS (per DU) - DEFAULT END-OF-LIFE SCENARIO

The default scenario represents recycling of all metals in the product and landfilling of all non-metallic constituents.
Environmental indicator results for all the declared modules are shown in the following tables for the declared unit of 1m^2 of Radii Planet Group's FirePro Ei120 single glazed fire rated partition system; Modules A1 - A3 are shown on an aggregated basis, as mandated by PCR2019:14.

The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (module A1-A3).
Environmental performance results are relative expressions and do not predict impacts on category endpoints, the exceeding of thresholds, safety margins or risks.

MANDATORY ENVIRONMENTAL PERFORMANCE INDICATORS ACCORDING TO EN 15804										
IMPACT CATEGORY	UNIT	A1 - A3	A4	A5	C1	C2	C3	C4	D	Modules A-C best product result variation %[1]
GWP-total	kg CO_2-eq	2.50E+02	7.72E+00	4.92E+00	3.92E-02	1.82E+00	1.07E-01	1.22E+00	-9.18E+00	95%
GWP-fossil	kg CO_2-eq	2.51E+02	7.71E+00	9.04E-02	3.90E-02	1.82E+00	1.07E-01	1.21E+00	-9.18E+00	96%
GWP-biogenic	kg CO_2-eq	-1.13E+00	5.28E-03	4.83E+00	7.38E-05	1.25E-03	5.51E-05	9.71E-03	0.00E+00	89%
GWP-luluc	kg CO_2-eq	2.13E-01	2.52E-03	6.01E-05	4.76E-05	5.94E-04	3.33E-05	6.54E-04	-1.06E-03	84%
ODP	kg CFC11-eq	4.86E-06	1.53E-07	2.55E-09	1.98E-09	3.61E-08	2.39E-09	3.09E-08	1.03E-07	110%
AP	mol H^+ eq	8.79E-01	2.41E-02	6.77E-04	1.45E-04	5.69E-03	8.27E-04	8.32E-03	-4.62E-02	83%
EP-freshwater	kg P eq	7.08E-02	5.14E-04	2.93E-05	7.02E-06	1.21E-04	4.50E-06	1.19E-04	-4.33E-03	76%
EP-marine	kg N eq	1.51E-01	8.13E-03	3.14E-04	3.14E-05	1.92E-03	3.73E-04	3.39E-03	-1.03E-02	88%
EP-terrestrial	mol N eq	1.75E+00	8.84E-02	3.08E-03	3.63E-04	2.09E-02	4.09E-03	3.44E-02	-1.05E-01	91%
POCP	kg NMVOC eq	6.76E-01	3.78E-02	7.84E-04	1.01E-04	8.91E-03	1.22E-03	1.19E-02	-3.41E-02	90%
ADPMM	kg Sb-eq	2.27E-03	2.52E-05	7.66E-07	6.59E-07	5.95E-06	8.66E-08	2.68E-06	-8.27E-05	106%
ADPFF	MJ	3.76E+03	1.08E+02	1.51E+00	1.06E+00	2.55E+01	1.71E+00	2.65E+01	-5.13E+01	98%
WDP	m^3	5.90E+01	5.23E-01	1.18E-01	1.04E-02	1.23E-01	7.64E-03	1.20E-01	-2.50E+00	84%

ADDITIONAL MANDATORY ENVIRONMENTAL PERFORMANCE INDICATOR										
IMPACT CATEGORY	UNIT	A1 - A3	A4	A5	C1	C2	C3	C4	D	Modules A-C best product result variation % [1]
GWP-GHG	kg CO_2-eq	2.51E+02	7.71E+00	9.05E-02	3.91E-02	1.82E+00	1.07E-01	1.22E+00	-9.19E+00	96%

[1] The A-C value for the best-case product is the stated percentage of the A-C value for the main product represented in the EPD

RESOURCE USE INDICATORS ACCORDING TO EN 15804

IMPACT CATEGORY	UNIT	A1 - A3	A4	A5	C1	C2	C3	C4	D
PERE	MJ. ncv	3.37E+02	1.84E+00	5.32E+01	3.55E-01	4.33E-01	1.82E-01	3.67E-01	1.51E+00
PERM	MJ. ncv	5.28E+01	0.00E+00	-5.28E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ. ncv	3.90E+02	1.84E+00	3.66E-01	3.55E-01	4.33E-01	1.82E-01	3.67E-01	1.51E+00
PENRE	MJ. ncv	3.76E+03	1.08E+02	1.51E+00	1.06E+00	2.55E+01	1.71E+00	2.65E+01	-5.14E+01
PENRM	MJ. ncv	1.11E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ. ncv	3.76E+03	1.08E+02	1.51E+00	1.06E+00	2.55E+01	1.71E+00	2.65E+01	-5.14E+01
SM	kg	1.70E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.13E+00
RSF	MJ. ncv	1.09E+01	3.42E-02	4.04E-04	3.19E-04	8.06E-03	2.45E-04	3.38E-03	2.54E-01
NRSF	MJ. ncv	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	2.83E+00	1.44E-02	9.09E-04	2.48E-04	3.39E-03	1.87E-04	-3.87E-01	3.40E-02

WASTE INDICATORS ACCORDING TO EN 15804

IMPACT CATEGORY	UNIT	A1 - A3	A4	A5	C1	C2	C3	C4	D
HWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NHWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
TRWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

OUTPUT FLOW INDICATORS ACCORDING TO EN 15804

IMPACT CATEGORY	UNIT	A1 - A3	A4	A5	C1	C2	C3	C4	D
CFR	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	1.28E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.51E+00	0.00E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	6.30E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	2.46E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

INDICATOR RESULTS (per DU) - ALTERNATIVE END-OF-LIFE SCENARIOS:

MANDATORY ENVIRONMENTAL PERFORMANCE INDICATORS ACCORDING TO EN 15804

IMPACT CATEGORY	UNIT	100% LANDFILL					100% RECYCLING				
		C1	C2	C3	C4	D	C1	C2	C3	C4	D
GWP-total	kg CO_2-eq	3.92E-02	1.82E+00	0.00E+00	1.27E+00	2.91E+00	3.92E-02	1.82E+00	9.99E-01	3.50E-02	-7.59E+01
GWP-fossil	kg CO_2-eq	3.90E-02	1.82E+00	0.00E+00	1.26E+00	2.91E+00	3.90E-02	1.82E+00	9.93E-01	3.50E-02	-7.58E+01
GWP-biogenic	kg CO_2-eq	7.38E-05	1.25E-03	0.00E+00	9.72E-03	0.00E+00	7.38E-05	1.25E-03	5.92E-03	1.28E-05	0.00E+00
GWP-luluc	kg CO_2-eq	4.76E-05	5.94E-04	0.00E+00	6.82E-04	1.09E-03	4.76E-05	5.94E-04	6.52E-04	1.39E-05	-4.54E-02
ODP	kg CFC11-eq	1.98E-09	3.61E-08	0.00E+00	3.25E-08	1.31E-07	1.98E-09	3.61E-08	2.21E-08	7.70E-10	-1.48E-06
AP	mol H^+ eq	1.45E-04	5.69E-03	0.00E+00	8.70E-03	3.50E-03	1.45E-04	5.69E-03	6.46E-03	1.90E-04	-6.40E-01
EP-freshwater	kg P eq	7.02E-06	1.21E-04	0.00E+00	1.23E-04	1.10E-04	7.02E-06	1.21E-04	4.67E-04	2.26E-06	-1.38E-02
EP-marine	kg N eq	3.14E-05	1.92E-03	0.00E+00	3.53E-03	1.09E-03	3.14E-05	1.92E-03	2.16E-03	2.80E-04	-9.71E-02
EP-terrestrial	mol N eq	3.63E-04	2.09E-02	0.00E+00	3.60E-02	1.20E-02	3.63E-04	2.09E-02	2.33E-02	7.90E-04	-1.10E+00
POCP	kg NMVOC eq	1.01E-04	8.91E-03	0.00E+00	1.25E-02	5.55E-03	1.01E-04	8.91E-03	7.92E-03	2.84E-04	-3.57E-01
ADPMM	kg Sb-eq	6.59E-07	5.95E-06	0.00E+00	2.77E-06	3.01E-06	6.59E-07	5.95E-06	3.85E-06	4.35E-08	-4.46E-04
ADPFF	MJ	1.06E+00	2.55E+01	0.00E+00	2.78E+01	5.55E+01	1.06E+00	2.55E+01	1.72E+01	6.54E-01	-9.77E+02
WDP	m^3	1.04E-02	1.23E-01	0.00E+00	1.24E-01	2.19E-01	1.04E-02	1.23E-01	1.54E-01	1.88E-03	-1.67E+01

ADDITIONAL MANDATORY ENVIRONMENTAL PERFORMANCE INDICATOR

IMPACT CATEGORY	UNIT	100% LANDFILL					100% RECYCLING				
		C1	C2	C3	C4	D	C1	C2	C3	C4	D
GWP-GHG	kg CO_2-eq	3.91E-02	1.82E+00	0.00E+00	1.27E+00	2.92E+00	3.91E-02	1.82E+00	9.94E-01	3.50E-02	-7.62E+01

RESOURCE USE INDICATORS ACCORDING TO EN 15804

IMPACT CATEGORY	UNIT	100% LANDFILL					100% RECYCLING				
		C1	C2	C3	C4	D	C1	C2	C3	C4	D
PERE	MJ. ncv	3.55E-01	4.33E-01	0.00E+00	3.79E-01	6.44E+00	3.55E-01	4.33E-01	1.30E+00	6.24E-03	-1.35E+02
PERM	MJ. ncv	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ. ncv	3.55E-01	4.33E-01	0.00E+00	3.79E-01	6.44E+00	3.55E-01	4.33E-01	1.30E+00	6.24E-03	-1.35E+02
PENRE	MJ. ncv	1.06E+00	2.55E+01	0.00E+00	2.78E+01	5.55E+01	1.06E+00	2.55E+01	1.72E+01	6.54E-01	-9.77E+02
PENRM	MJ. ncv	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ. ncv	1.06E+00	2.55E+01	0.00E+00	2.78E+01	5.55E+01	1.06E+00	2.55E+01	1.72E+01	6.54E-01	-9.77E+02
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.66E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	-4.25E+01
RSF	MJ. ncv	3.19E-04	8.06E-03	0.00E+00	3.50E-03	1.01E-02	3.19E-04	8.06E-03	2.91E-02	6.42E-05	-6.07E-01
NRSF	MJ. ncv	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	2.48E-04	3.39E-03	0.00E+00	-3.86E-01	5.33E-03	2.48E-04	3.39E-03	-4.74E-02	2.82E-04	-3.56E-01

WASTE INDICATORS ACCORDING TO EN 15804

IMPACT CATEGORY	UNIT	100% LANDFILL					100% RECYCLING				
		C1	C2	C3	C4	D	C1	C2	C3	C4	D
HWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NHWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
TRWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

OUTPUT FLOW INDICATORS ACCORDING TO EN 15804											
IMPACT CATEGORY	UNIT	100% LANDFILL					100% RECYCLING				
		C1	C2	C3	C4	D	C1	C2	C3	C4	D
CFR	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.17E+02	0.00E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

INTERPRETATION OF THE LCA RESULTS

Glass manufacture makes the largest contribution to the indicator values in Modules A1-A3. Powder coating contributes approximately 4% of the GWPtotal in Modules A1-A3.

Indicator values obtained for resource depletion (ADPMM, ADPFF), stratospheric ozone depletion (ODP), eutrophication (EP) freshwater and water deprivation potential (WDP) should be used with caution; all are subject to uncertainties in data or method which limit the scope for their use as the basis for comparisons.

No untreated wastes leave the modelled system, which includes waste treatment activities as required by EN 15804. Therefore all waste indicators are zero; very long term managed storage is interpreted as waste treatment.

The secondary materials (SM) indicator does not represent the recycled material content of the product.

The output flow indicators also include flows leaving the background system.

The reporting of Module D shows benefits as negative indicator values.

ADDITIONAL INFORMATION

ENVIRONMENTAL MANAGEMENT, ECONOMIC AND SOCIAL INFORMATION

Radii Planet Group recognises and is acting upon its legal and moral responsibility to manage its business operations and reduce any detrimental impact on the environment.

The company is committed to ensuring environmental best practise in line with its environmental and sustainability policies and in compliance with ISO 14001:2015. This commitment covers multiple parts of the business operations, including:

Design

- *systems are designed and drawn in order to minimise waste. Projects are extensively surveyed and CAD detailed. This allows for great accuracy in material take off and efficient calibration.*

Responsible Sourcing of Materials

- *sourcing materials from ethical and sustainable sources and providing alternatives where Radii Planet Group believe sustainability could be better achieved;*
- *ensuring all timber materials used in projects are FSC-certified and, where required by its clients, providing a valid 'chain of custody' as part of its commitment to dependable supply.*

Local Economies and Community Enrichment

- *where possible, using local suppliers for materials and labour, including SMEs and minority owned businesses, to improve local economies and reduce carbon emissions through the reduction of transportation distances;*
- *reducing its impact on air quality through the careful selection and maintenance of plant and equipment;*
- *controlling work operations in order to reduce noise and disruption caused by its activities;*

Carbon Reduction

- *In line with its Carbon Reduction Policy, putting in place carbon reduction measures to eliminate carbon emissions generated by its activities, including energy use, embodied carbon, business travel, waste and water.*

ABBREVIATIONS

DU: declared unit

EWC: European Waste Code

FSC: Forest Stewardship Council®: a certification scheme for timber procurement (https://fsc.org/en)

IES: The International EPD® System: a programme for Type III environmental declarations, maintaining a system to verify and register EPD®s as well as keeping a library of EPD®s and PCRs in accordance with ISO 14025. (www.environdec.com)

LCA: Life cycle assessment: LCA studies the environmental aspects and quantifies the potential impacts (positive or negative) of a product (or service) throughout its entire life. ISO standards ISO 14040 and ISO 14044 set out conventions for conducting LCA.

NCV: net calorific value

PCR: Product Category Rules

REACH Regulation: REACH is the European Regulation on Registration, Evaluation, Authorisation and Restriction of Chemicals. It entered into force in 2007, replacing the former legislative framework for chemicals in the EU.

UN CPC: UN (United Nations) Central Product Classification

REFERENCES

Certifire - independent third-party passive fire product certification scheme

https://www.warringtonfire.com/certification-services/fire-certification/certifire

ecoinvent database (v3.10) - www.ecoinvent.ch

EF3.1 Characterisation Factors for Use with EN 15804
https://eplca.jrc.ec.europa.eu/LCDN/developerEF.xhtml

EN 15804:2012 + A2:2019 - Sustainability of construction works - Environmental Product Declarations - Core rules for the product category of construction products

FirePro Partitions & doors: LCA & EPD - Report for Radii Planet Group Ltd (2025) - EuGeos Limited

General Program Instructions (GPI) Version 5.0.0 (2024-06-19) - The International EPD® System - EPD International AB

ISO 9001:2015 - Quality management system. Requirements

ISO 14001:2015 - Environmental management systems – Requirements with guidance for use.

ISO 14025:2009-11: Environmental labels and declarations - Type III environmental declarations - Principles and procedures

ISO 45001:2018 - Occupational health and safety management systems — Requirements with guidance for use

openLCA - LCA software developed by GreenDelta since 2006 - https://www.openlca.org/

PCR 2019:14 Construction products (EN 15804+A2), V2.0.1 - 2025-06-05 - EPD International AB

Spindler C, Ciroth A, openLCA EN 15804 add-on, report v2.1, June 19, 2023
available at https://nexus.openlca.org/database/EN15804%20add-on

VERSION HISTORY

V001 – 2026-03-09- Original version of the EPD